

Quan Xiao | Curriculum Vitae

1 E Loop Rd, New York City, NY, 10044

🌐 Homepage
in LinkedIn

✉ quanx1808@gmail.com
🔗 Google Scholar

✉ qx232@cornell.edu
🔗 JenniferquanXiao

EDUCATION

Cornell University (Cornell Tech)

New York, NY

Ph.D. of Electrical and Computer Engineering (GPA: 4.3/4.3)

Aug. 2025 - present

Advisor: Tianyi Chen

Rensselaer Polytechnic Institute

Troy, NY

Ph.D. of Computer and Systems Engineering (GPA: 4.0/4.0)

Sep. 2021 - (Transferred) Aug. 2025

Advisor: Tianyi Chen

University of Science and Technology of China

Hefei, Anhui, China

Bachelor of Statistics

Sep. 2016 - July. 2020

RESEARCH INTEREST

My research focuses on optimization and machine learning, particularly **bilevel and multi-objective optimization**, to develop 1) **multi-goal-oriented methods for generative models**, including LLMs and diffusion models; 2) while explicitly accounting for physical and system-level constraints of hardware platforms, including **analog in-memory computing (AIMC) for energy-efficient AI training and deployment**. More broadly, I aim to bridge theory and practice by designing scalable and principled methods for safe, reliable, efficient, and hardware-friendly AI systems.

HONORS AND AWARDS

ICML Gold Reviewer	2026
Travel Award for NYC Ops Day	2026
NeurIPS Top Reviewer	2025
IEEE Signal Processing Society Scholarship	2024
Belsky Award for Computational Sciences and Engineering in RPI	2023
Neural Information Processing Systems Scholar Award	2023
International Conference on Machine Learning Participation Grant	2022
Successful Participant in Mathematical Contest In Modeling	2019
Excellent student scholarship of the University of Science and Technology of China	2017 - 2019

SELECTED PUBLICATIONS (* equal contribution)

- [13] ***Quan Xiao**, *Jindan Li, Zhaoxian Wu, Tayfun Gokmen, and Tianyi Chen. Dynamic Symmetric Point Tracking: Tackling Non-ideal Reference in Analog In-memory Training. *Proc. of International Conference on Machine Learning*, 2026.
- [12] Zhaoxian Wu, **Quan Xiao**, Tayfun Gokmen, Hsinyu Tsai, Kaoutar El Maghraoui, Tianyi Chen. On the Convergence Theory of Pipeline Gradient-based Analog In-memory Training. *IEEE Journal on Selected Areas in Information Theory*, 2026.
- [11] Zhaoxian Wu, **Quan Xiao**, Tayfun Gokmen, Omobayode Fagbohunge, and Tianyi Chen. Analog In-memory Training on General Non-ideal Resistive Elements: Understanding the Impact of Response Functions. *Proc. of Neural Information Processing Systems*, 2025. (**Oral, top 0.3%**)

- [10] Liuyuan Jiang, **Quan Xiao**, Lisha Chen, and Tianyi Chen. Beyond value functions: Single-loop bilevel optimization under flatness conditions. *Proc. of Neural Information Processing Systems*, 2025.
- [9] **Quan Xiao**, Hui Yuan, A F M Saif, Gaowen Liu, Ramana Rao Kompella, Mengdi Wang, Tianyi Chen. A First-order Generative Bilevel Optimization Framework for Diffusion Models. *Proc. of International Conference on Machine Learning*, 2025.
- [8] Lisha Chen, **Quan Xiao**, Ellen Hidemi Fukuda, Xinyi Chen, Kun Yuan, Tianyi Chen. Efficient Multi-Objective Learning under Preference Guidance: A First-Order Penalty Approach. *Proc. of International Conference on Machine Learning*, 2025. **(Spotlight, top 2.6%)**
- [7] **Quan Xiao**, and Tianyi Chen. Unlocking Global Optimality in Bilevel Optimization: A Pilot Study. *Proc. of International Conference on Learning Representations*, 2025.
- [6] Han Shen, **Quan Xiao**, and Tianyi Chen. On Penalty-based Bilevel Gradient Descent Method. *Mathematical Programming (Series A)*, 2024.
- [5] Liuyuan Jiang, **Quan Xiao**, Victor M Tenorio, Fernando Real-Rojas, Antonio Marques, Tianyi Chen. A Primal-Dual-Assisted Penalty Approach to Bilevel Optimization with Coupled Constraints. *Proc. of Neural Information Processing Systems*, 2024
- [4] **Quan Xiao**, Songtao Lu, and Tianyi Chen. A Generalized Alternating Method for Bilevel Learning under the Polyak-Łojasiewicz Condition. *Proc. of Neural Information Processing Systems*, 2023.
- [3] **Quan Xiao**, Han Shen, Wotao Yin and Tianyi Chen. Alternating Projected SGD for Equality-constrained Bilevel Optimization. *Proc. of International Conference on Artificial Intelligence and Statistics*, 2023.
- [2] *Momin Abbas, ***Quan Xiao**, *Lisha Chen, Pin-Yu Chen and Tianyi Chen. Sharp-MAML: Sharpness-Aware Model-Agnostic Meta Learning. *Proc. of International Conference on Machine Learning*, 2022.
- [1] Tianyi Chen, Yuejiao Sun, **Quan Xiao**, and Wotao Yin. A Single-Timescale method for stochastic bilevel optimization. *Proc. of International Conference on Artificial Intelligence and Statistics*, 2022. **(Oral, top 2%)**

INVITED TALK

- **Unlocking Nested Learning for AI: A Case Study on Data Selection for LLM.**
 NYC Operations (Ops) Day, New York, NY. Apr. 2026
 Information Theory and Applications (ITA) Workshop, San Diego, CA. Feb. 2026
- **Global Convergence in Bilevel Optimization: A Pilot Study.**
 Conference on Parsimony and Learning (CPAL), Stanford, CA. March. 2025
 Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA. Nov. 2024
- **Efficient Methods for Bilevel Optimization with Polyak-Łojasiewicz Lower-level Problems.**
 Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA. Nov. 2023
 SIAM Conference on Optimization (OP23), Seattle, WA. June. 2023

INTERNSHIP

- **IBM Thomas J. Watson Research Center**, Yorktown, NY.
 Research intern with mentor: Dr. Debarun Bhattacharjya. May. 2024 - Aug. 2024
Project: Uncertainty quantification and selective generation in large language model.
- **IBM Thomas J. Watson Research Center**, Yorktown, NY.
 Research intern with mentor: Dr. Songtao Lu. May. 2022 - Aug. 2022
Project: Efficient method for bilevel optimization beyond strong convexity.

SERVICES

- Session Co-Chair at the Asilomar Conference on Signals, Systems, and Computers 2024
- Reviewer of Proc. of AAAI ('25), AISTATS ('22 – '26), NeurIPS ('23 – '26), ICLR ('24 – '26), ICML ('24 – '26).
- Reviewer of journal of IEEE Transactions on Signal Processing, IEEE Transactions on Networking.
- Reviewer and Technical Program Committee member of Cross-Community Federated Learning Workshop at MLSys 2022.